Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20191001-027-R0 (signed scan)	Issued	2019-10-18

UT Thickness Re-baseline Survey - ATU-100 (signed)

Scope and Objectives

Larkhall NDE Services Inc. was engaged to perform a comprehensive re-baseline ultrasonic thickness survey of the ATU-100 at the Hamilton Plant. This survey was executed to align the unit with the updated QSCP-CIS-001 Rev 0, which mandates a re-baseline of all Condition Monitoring Locations (CMLs) to support Risk-Based Inspection (RBI) data requirements. The scope encompasses 150 CMLs distributed across 18 process lines, providing a new data foundation for future Fitness-For-Service (FFS) assessments by qualified engineers.

The inspection program was governed by QSCP-IP-ATU100 Rev 3, which rationalized the CML count and established revised inspection intervals. Per the RBI strategy, Class 1 sour gas, regenerator overhead, and rich amine lines are now subject to 3-year intervals, while the remaining lines are set at 6-year intervals. This report serves as the primary record for this re-baseline event, with a scanned signed copy filed alongside the native document to ensure data integrity.

Personnel and Procedures

The survey was performed under QSCP-NDT-001 Rev 2, which incorporates specific requirements for the CUI program, including insulation removal and the use of inspection ports. All NDE activities were conducted in accordance with API 570 and ASTM E797. The inspection team consisted of Trevor Nolan, acting as the primary author and UT Level II technician, and Julia Martinez, who provided UT Level II and RT Level II technical support. The qualifications of the crew are maintained in accordance with ASNT SNT-TC-1A.

Quality oversight and technical review were provided by David Park, serving as the Level III and Quality Manager. All measurements were obtained using an Olympus 37DL PLUS 0611452. The instrument was verified against LNDE-CAL-2019-0588 prior to the commencement of the survey. For the stainless steel reflux lines, the gauge was calibrated to the specific velocity settings required for A312 TP316 material to ensure measurement accuracy.

Process Piping - Gas and Amine Services

High-pressure gas lines 6-SG-101 and 6-SG-102 (A106 Gr.B, 168.3 mm OD) were inspected against a t-min of 3.56 mm. The treated gas line 8-TG-101 was also surveyed, noting a design pressure of 1896 kPa(g). Lean amine supply lines 4-LA-101, 4-LA-102, and the header 6-LA-103 were evaluated against their respective design temperatures of 93 °C. The lean side of the L/R exchanger, 4-LA-104, was also included in this circuit.

Rich amine lines 4-RA-101, 4-RA-102, and the header 6-RA-103 were prioritized due to the 3-year RBI interval. These lines, along with the rich side of the L/R exchanger 4-RA-104, were compared against historical data from LNDE-RPT-20140902-024-R0, which was the first survey performed by Larkhall NDE Services Inc. following the 2011-02 readings. All A106 Gr.B components were assessed for corrosion rates relative to their nominal wall thicknesses.

Overhead, Reflux, and Utility Systems

Regenerator overhead lines 3-RV-101 and 2-RV-102 were inspected as Class 1 systems. These lines have a design temperature of 121 °C and were evaluated against t-min values of 3.56 mm and 2.79 mm respectively. The stainless steel reflux lines 3-RF-101 and 2-RF-102 were measured using the appropriate velocity settings for A312 TP316. These lines form part of the critical overhead circuit defined in the piping inspection plan.

Utility services including reboiler steam line 2-ST-101 and amine cooler cooling water lines 3-CW-101 and 3-CW-102 were surveyed as Class 3 systems. These lines are subject to 10-year code maximum intervals, though the current RBI plan sets a 6-year frequency. All measurements were recorded to establish the new baseline for short-term and long-term corrosion rate calculations as required by the corporate integrity standard.

Data Analysis and Compliance

The results of this survey were compared with the previous full-unit survey recorded in LNDE-RPT-20160118-004-R0. Corrosion rates were calculated as the larger of the long-term rate and the short-term rate over the latest pair of readings at least 3 years apart. All CML readings were found to be above the retirement thickness (t-min) specified in the line facts. No Non-Conformance Reports (NCRs) were issued during this re-baseline event.

Remaining life calculations were performed using the formula (t - t-min) / rate. In accordance with site rules and ASME B31.3 design considerations, the next inspection dates have been scheduled at half the remaining life, not to exceed the class maxima or the 3/6 year RBI intervals. The data collected during this survey meets the requirements for the CUI program, ensuring that insulated lines in the CUI range are monitored via the established profile radiography and insulation window protocols.

Sign-off	Name	Function	Date	Signature
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level II	2019-10-18	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2019-10-18	_____